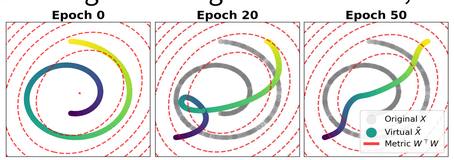
The Weight Gram Matrix Captures Sequential Feature Linearization in Deep Networks

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TL;DR Gradient descent implicitly learns the covariance of *virtually* updated features through the weight Gram matrix, driving deep networks to *sequentially linearize representations* toward the target, layer by layer.



Feature-Centric Framework

Previous works	Our work
Focus on weight dynamics, $\nabla_W \mathcal{L}$ Degree of movement (rich & lazy) Degree of alignment (t-s setting)	Analyze feature dynamics through $\nabla_h \mathcal{L}$

How are they related?

For any function f that takes $z = Wh$ as inputs,

$\nabla_W f = \nabla_z f \; h^\top$ and $\nabla_h f = W^\top \nabla_z f$

$$\Rightarrow W^\top \nabla_W f = \nabla_h f \; h^\top$$

$\nabla_W f$: how the weight would move to optimize f
$\nabla_h f$: how the hidden state would move to optimize f

When does this relationship appear?

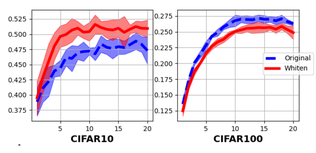
$$(W - \gamma \nabla_W \mathcal{L})^\top (W - \gamma \nabla_W \mathcal{L}) = W^\top W - \gamma W^\top \nabla_W \mathcal{L} + \dots$$
$$= W^\top W - \gamma \nabla_h \mathcal{L} \; h^\top + \dots$$

It appears under the weight Gram structure!

Role of Gram in feature learning

Whitening the Gram-invariant parts

$$\arg\min_\Delta \|\Delta\|_F \; s.t.$$
$$(W - \gamma\Delta)^\top (W - \gamma\Delta) =$$
$$(W - \gamma \nabla_W \mathcal{L})^\top (W - \gamma \nabla_W \mathcal{L})$$



Virtual Covariance Learning

Definition.

- l-th hidden state: $h_l = \sigma(W_{l-1} h_{l-1})$
- **Actual update** (weight): $W^+ = W - \gamma \nabla_W \mathcal{L}$
- **Actual update** (hidden): $\sigma(W_{l-1}^+ h_{l-1})$
- **Virtual update**: $h^+ = h - \gamma \nabla_h \mathcal{L}$
- **Virtual (uncentered) covariance shift**: $h^+ (h^+)^\top - h h^\top$

Thm. 1 (GD is Virtual Covariance Learning)

$$(W^+)^\top W^+ - W^\top W \approx h^+ (h^+)^\top - h h^\top$$

GD updated weight Gram learns the covariance structure of Virtually updated hidden states

Prop. 1 & 2 (relationship b.w. actual & virtual updates)

- Under single-sample training and gradient flow regime,

$$\frac{dh}{dt} = -M \nabla_h \mathcal{L}, \text{ for p.s.d. } M$$

Actual update = preconditioned Virtual update

- Under single-sample/single-layer training,

$$|\sigma(W_{l-1}^+ h_{l-1}) - \sigma(W_{l-1} h_{l-1})| \le c^2 |h_l^+ - h_l| \text{ and}$$
$$sgn(\sigma(W_{l-1}^+ h_{l-1}) - \sigma(W_{l-1} h_{l-1})) = sgn(h_l^+ - h_l)$$

First-Order Approximation in Sum

Prop. 5 (first-order approximation can be exact)

For a homogeneous function f with centered input $\bar{x} = 0$ and prediction $\bar{f} = 0$,

$$\underbrace{\Sigma_{i,j}}_{\text{Sum of errors}} \Big[\underbrace{f(x_i)}_{i\text{-th prediction}} - \underbrace{\big(f(x_j) + (x_i - x_j)^\top \nabla_{x_j} f \big)}_{\text{Taylor approximation from j-th prediction}} \Big] = 0$$

Brute-force approximation can be exact in aggregate

Sequential Feature Linearization

Definition.

- **Target Linearity (TL)**: $1 - \frac{\|Y - \hat{Y}_\lambda\|_2^2}{\|Y - \bar{Y}\|_2^2}$
  - Y: Prediction label, $\bar{Y}$: its mean, H: hidden states
  - $\hat{Y}_\lambda = H^\top (HH^\top + \lambda I)^{-1} HY$ (Ridge regression prediction)
    R^2 *score of Ridge regressor*
- **Surrogate**: $Y^\top H^\top \boldsymbol{W}^\top \boldsymbol{W} HY$
  - W: weight matrix corresponding to H
    Linear kernel alignment between Y *and* WH
  - **Thm. 2:** Target Linearity $\ge 1 - C(\text{Surrogate})^{-1}$

Thm. 3 (Training dynamics)

$$\text{Surrogate}_{t+1} - \text{Surrogate}_t \approx 2\gamma \underbrace{(f^\top y)}_{\substack{\text{Corr. between pred. and label,} \\ 0 \to \|y\|_2^2}} \cdot \underbrace{(y^\top K g)}_{\substack{K\text{-induced corr. between} \\ \text{grad. and label}}}$$

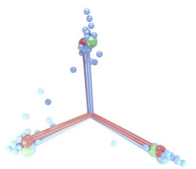
f: prediction, y: label, $g = -\nabla_f \mathcal{L}$, $K_{ij} = h_i^\top h_j$

- Under BCE loss, $y^\top K g = \langle \text{pos. hiddens}, \text{pos.} - \text{neg. hiddens} \rangle$

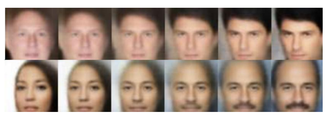
Thm. 4 (Depth dynamics)

Under the Neural Feature Ansatz, when NN interpolates, the lower bound of Surrogate increases w.r.t. layer index

Other Feature Learning Phenomena



Prop. 3
Neural Collapse is an extreme case of TL maximization



Prop. 4
High TL is a suf. condition for latent space linear interpolation

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